Metastable states and tipping in the Bayesian swarm

a comparison with "Indirect tipping" (Flint et al., arXiv:2609.25194)

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Abstract

Flint et al. study populations of LLM agents playing a coordination game. They map how large a committed minority must be to tip the population from one consensus to another. Their key notions are metastable consensus states, critical mass, an asymmetric transition landscape, stepping stones, and "unintended diversion" into a third state.

We ask whether our Bayesian swarm has metastable states and tipping, using only the existing 15-agent runs (1,000 episodes per cell, four setups). We classify each episode's end state by its vote shares. The analysis is exploratory: the definitions were fixed before looking at the maps, but no predictions were registered.

- **Three end states, in a clean phase diagram:** mixed (no majority), A2 lock-in (the attack succeeds) and A0 lock-in (the correct answer, or in no-solution setups a backfire). The A2 region is a diagonal band in (q, ρ), the same diagonal as the reading reversal.
- **Genuine bistability under attack** in the symmetric setups. Attacked episodes split between A2 lock-in and recovery towards A0. In the most bistable cell (task003, $q = 8$, $\rho = 0.61$), 40% end with A2 ≥ 0.8 and 35% with A2 ≤ 0.2, while the silent arm has no A2 mode.
- **Durable tipping:** 86–90% of A2 lock-ins present when the controller stops are still there 100 time units later.
- **Unintended diversion:** in the no-solution setups at long memory, the false controller raises the chance of an *A0* lock-in by up to 0.70.

The metastable states of our system are *information states*, defined by which facts survive. They are not conventions sustained by a wish to agree.

1 The paper, and how its concepts map onto our system

Flint et al. simulate the *naming game*. Pairs of agents each pick a word and score only when the words match; every word is equally good. The paper's notions:

- **Conventions:** consensus on one word, metastable because everyone is rewarded for matching.
- **Committed agents:** a fraction p that always says a target word.
- **Critical mass** ω_{ij}: the smallest p that flips consensus on i to j. It is strongly asymmetric between $i \to j$ and $j \to i$, and the ω_{ij} form a directed landscape.
- **Stepping stones:** indirect paths $i \to k \to j$ can be cheaper than the direct one, or make it possible.
- **Transient commitment:** it can work, but stopping at the wrong time diverts the population to a third convention.

Table 1: Concepts compared.

	Flint et al. (naming game)	our Bayesian swarm
what agents want	to match the partner (coordination payoff)	the correct answer; no payoff for agreeing; votes never seen
metastable state	consensus on one word	the set of surviving facts, and the beliefs it implies
why it is stable	social reinforcement	fact extinction is permanent; support-based reposting recirculates supporting facts
attacker	committed agents, a fraction p of the population, permanent	one external controller posting true facts at rate 1, about 1/16 of posts, until $t = 30$
attack lever	p	reading q and memory ρ of the *targets*; the controller's budget and rate (not scanned here)
transient attack	a tested strategy	our standard design (persistence)
third state	a latent convention can capture the population	A0 lock-in caused by the false controller in no-solution setups

2 Definitions

For episode e at time t, let $s_{k,e}(t)$ be the share of voters voting Ak. The episode is

$$\text{in A}k\text{ lock-in if } s_{k,e}(t) \geq \tfrac{2}{3}, \qquad \text{otherwise mixed.} \tag{1}$$

The threshold $\frac{2}{3}$ was fixed before the maps. A stricter 0.8 gives the same picture (correlation 0.96 for tipping, 0.93 for diversion, over all cells). With P^{c} and P^{s} the fractions of episodes in a state in the false-controller and silent arms, we define

$$\text{tipping probability} \quad T_{A2}(t) = P^{\mathrm{c}}(\text{A2 lock-in}, t) - P^{\mathrm{s}}(\text{A2 lock-in}, t), \tag{2}$$

$$\text{diversion into A0} \quad V_{A0}(t) = P^{\mathrm{c}}(\text{A0 lock-in}, t) - P^{\mathrm{s}}(\text{A0 lock-in}, t), \tag{3}$$

$$\text{bistability index} \quad B = \min\big[P^{\mathrm{c}}(s_2 \leq 0.2),\, P^{\mathrm{c}}(s_2 \geq 0.8)\big] \text{ at } t = 130. \tag{4}$$

B is large only when attacked episodes split between both extremes. All maps use standard reading.

3 Results

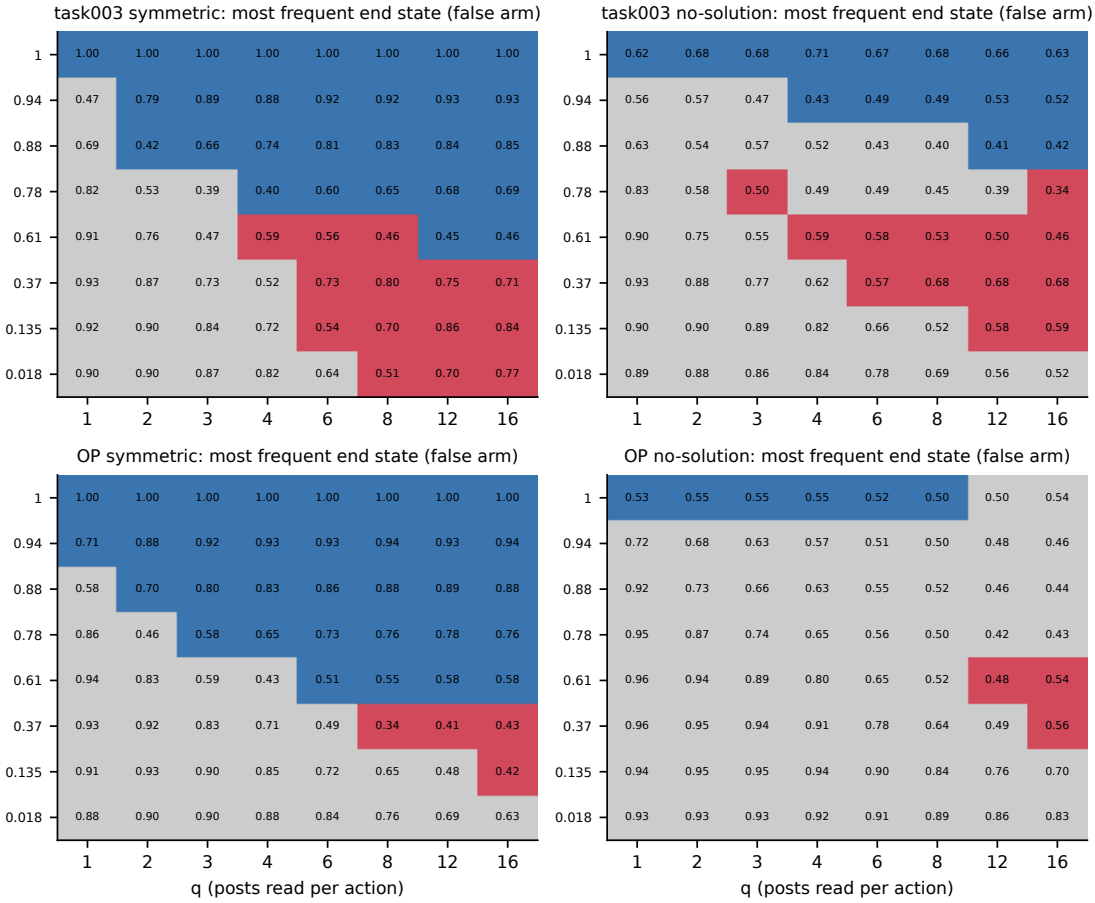


Figure 1: Phase diagram of end states, false-controller arm, $t = 130$ (100 time units after the stop). Colour: the most frequent state. Grey: mixed; red: A2 lock-in; blue: A0 lock-in. Number: its share of episodes.

Result 1 (three regions)**.** *In every setup the false arm's end states form three regions (Fig. 1):*

- ***mixed*** *at short memory and little reading;*
- ***A2 lock-in*** *in a diagonal band: the attack's success region;*
- ***A0 lock-in*** *at long memory.*

The band lies along the diagonal of constant $q\tau$ that also locates the reading reversal (campaign report of 30 October). Moving up in memory, the swarm passes from too forgetful to be captured *to* captured *to* informed enough to resist.

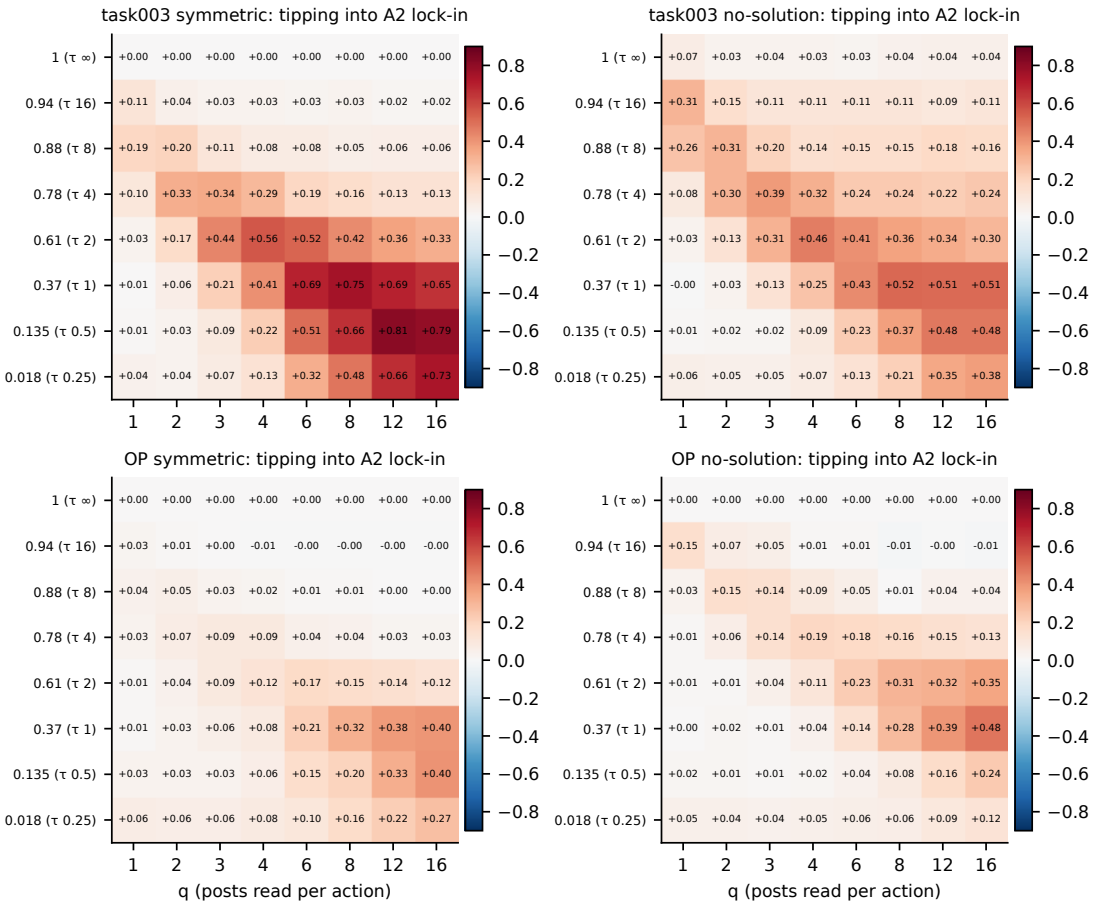


Figure 2: Tipping probability $T_{A2}(130)$, Eq. (2): the extra chance that an attacked episode is still locked into A2 100 time units after the attack. (The $t = 30$ version, `fig01_tipping_A2_t30.pdf`, is similar and slightly larger.)

Result 2 (tipping is strong and durable)**.** *The tipping probability reaches 0.87 at the stop and 0.81 a hundred time units later in task003 symmetric; 0.52–0.54 and 0.48–0.52 in the no-solution setups; 0.53 and 0.40 in OP symmetric. Of the A2 lock-ins present at the stop, the median share still present at* $t = 130$ *is:*

- *0.86 in task003 symmetric;*
- *0.90 in task003 no-solution;*
- *0.86 in OP no-solution;*
- *0.58 in OP symmetric.*

Once tipped, the swarm mostly stays tipped without the attacker. These are metastable states in the paper's sense.

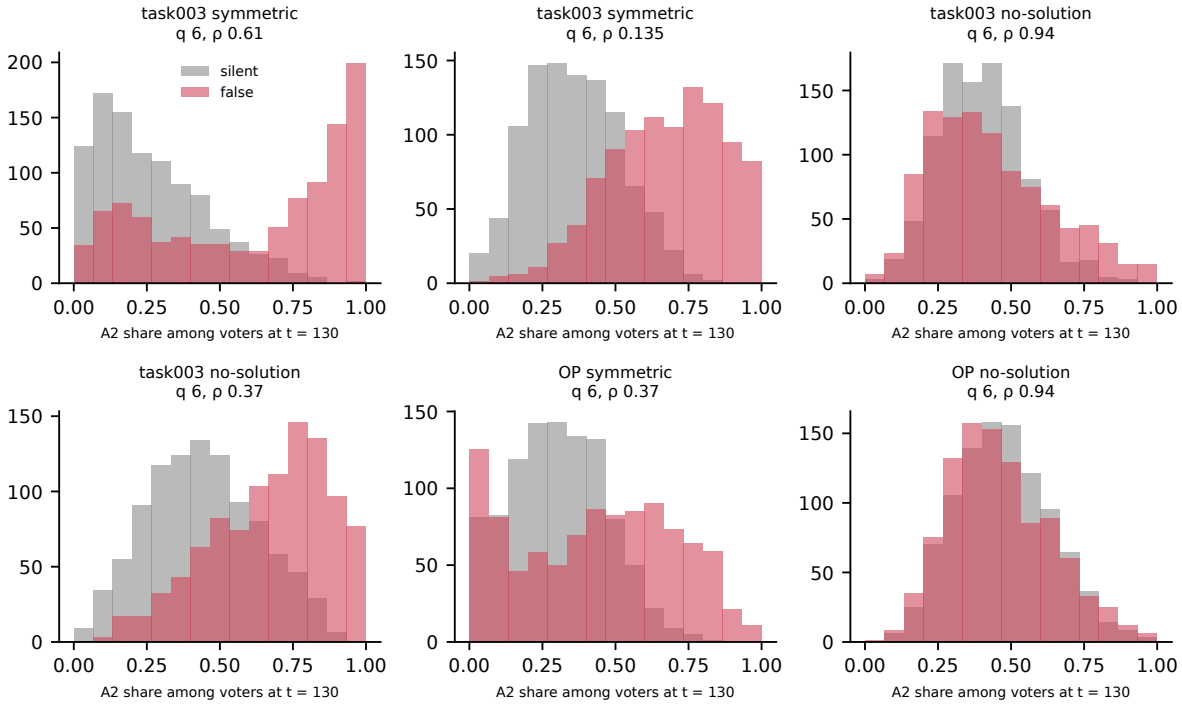


Figure 3: Per-episode A2 share at $t = 130$, silent (grey) against false arm (red), at selected cells.

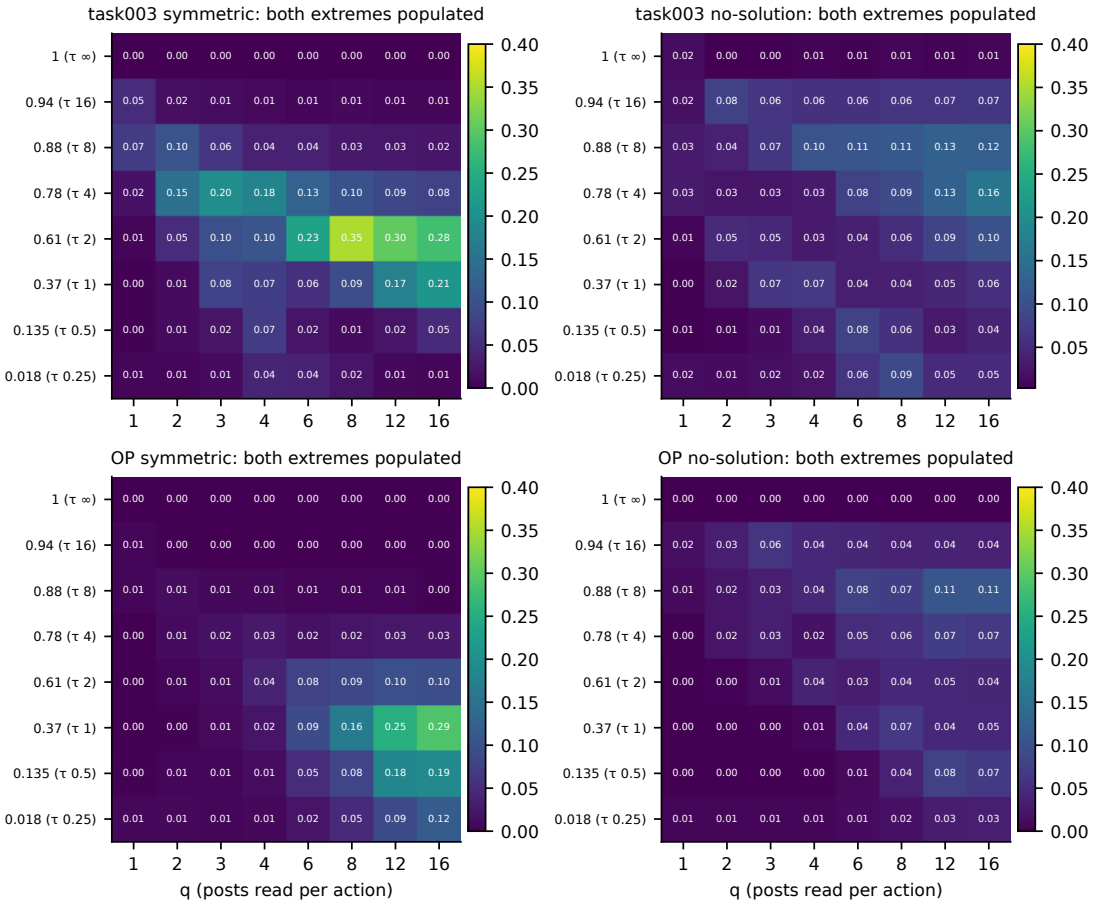


Figure 4: Bistability index B, Eq. (4).

Result 3 (bistability under attack, at the edge of the A2 band)**.** *In the symmetric setups, attacked episodes at the boundary between the A2 and A0 regions split into two outcomes (Figs. 3, 4).*

- ***task003 symmetric,*** $q = 8$, $\rho = 0.61$*: 40% of attacked episodes end with A2* ≥ 0.8 *and 35% with A2* ≤ 0.2*. In the silent arm only 1% reach A2* ≥ 0.8*.*
- ***OP symmetric,*** $q = 16$, $\rho = 0.37$*: 29% and 36%.*

This is the signature of two competing attractors: the correct answer the swarm can prove, and the A2 lock-in the attack creates. Which one an episode reaches is decided by its own fluctuations. The no-solution setups are barely bistable ($B \leq 0.16$*): the attack shifts the whole distribution without splitting it.*

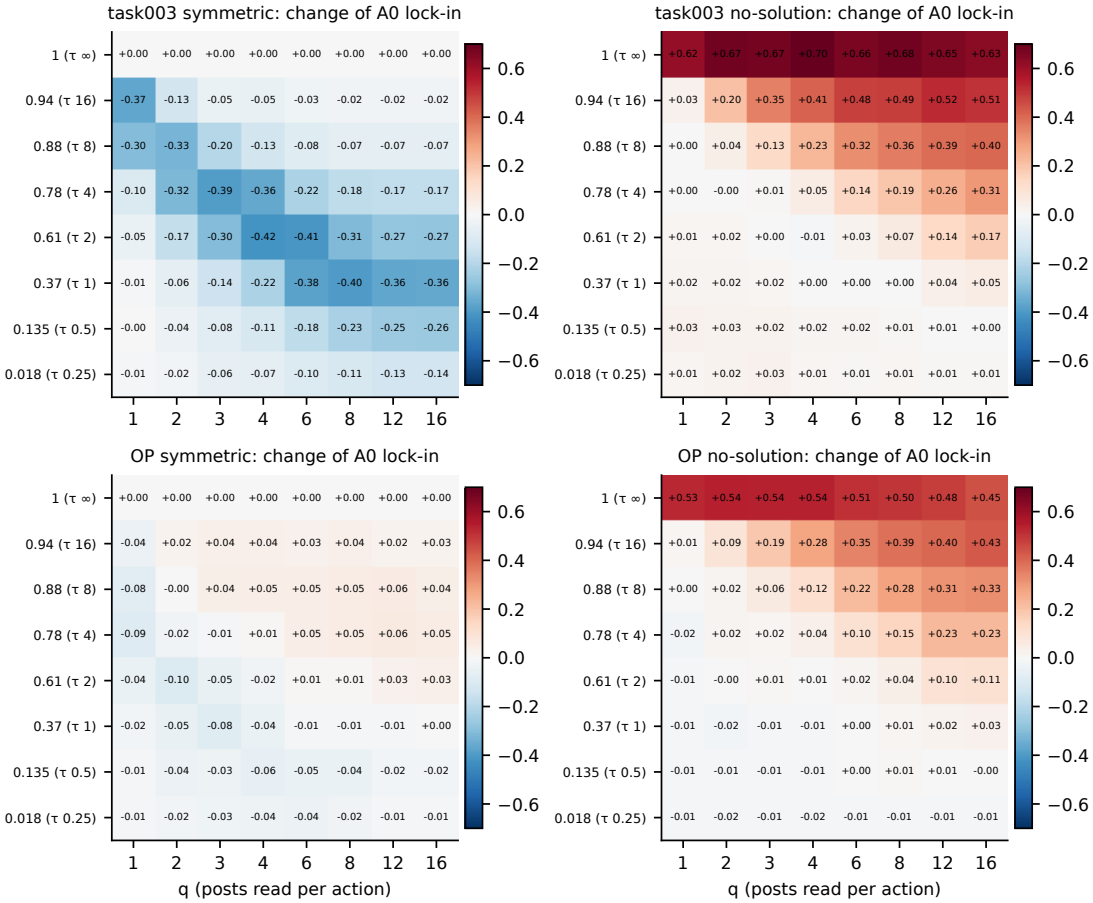


Figure 5: Diversion into A0 $V_{A0}(130)$, Eq. (3): the change in the chance of an A0 lock-in caused by the *false* controller.

Result 4 (the false controller can tip the swarm into the answer it opposes)**.** *In the no-solution setups at long memory, the false controller* raises *the chance of an A0 lock-in by up to 0.70 (task003) and 0.54 (OP), mostly at* $\rho \geq 0.88$ *(Fig. 5).*

- ***Why:*** *the controller's single pool, combined with the agents' own facts, gives* $(0.75, 0, 0.25)$ *in task003 and* $(0.67, 0, 0.33)$ *in OP. Long-memory agents who keep the controller's facts end up favouring A0.*
- ***The analogue in the paper:*** *its "unintended diversion", where a transient attack leaves the population on a latent third convention.*
- ***In the symmetric setups*** V_{A0} *is negative or zero: the attack only removes A0 lock-ins it would otherwise reach.*

4 Discussion: what carries over from the paper and what does not

- **Metastable states exist, but they are not three symmetric conventions.** Without an attacker, the no-solution swarm has no stable A0, A1 or A2 population. Probability-matching agents with near-uniform evidence vote close to at random, and A1 never wins anywhere. The long-lived states come from the information dynamics:
  - which facts survive, since an extinct fact never returns;

  - support-based reposting, which recirculates the facts behind the current majority.

  Our theory already contains this. The controller's facts persist when their growth number exceeds 1, and the mean-field model becomes bistable under a strong reposting advantage.

- **Critical mass has a natural analogue.** Here the lever was not the attacker's size, which was fixed, but the targets' reading and memory: the A2 band in Fig. 1 is our map of where the fixed attack exceeds its critical mass. The paper's question proper, the smallest controller budget or rate that tips a given (q, ρ), needs new runs scanning the controller.
- **Asymmetry and stepping stones** need several targets. Our runs have one false target (A2). Two direct tests:
  - a controller that targets A1 first and A2 later, at equal total budget, in the no-solution setups;
  - the reverse attack, truth $\rightarrow$ false against false $\rightarrow$ truth.
- **Transient attacks are our standard design.** The result that A2 lock-ins outlive the attacker matches the paper's observation that a population can acquire enough inertia to stay on the target after withdrawal.

Limits.

- One snapshot ($t = 130$) defines the end state; no first-passage or dwell-time analysis.
- The $\frac{2}{3}$ threshold is a convention, though robust against 0.8.
- Fifteen agents give coarse vote shares (steps of 1/15).

Files.

- `analysis/tipping.py`, which writes `table_tipping.csv`, in this folder.
- `analysis/tipping_figures.py`, which writes `figures/`.
- The paper: A. Flint, L. M. Aiello, S. M. Constantino, R. Pastor-Satorras, A. Baronchelli, "Indirect tipping: a social attack surface in AI agent populations", arXiv:2609.25194 (2026).